HASNet: A Foreground Association-Driven Siamese Network With Hard Sample Optimization for Remote Sensing Image Change Detection

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***Abstract*— Remote sensing change detection (RS-CD) relies on the model's ability to learn features of marked change objects, known as foreground targets. Beyond foreground targets, the background targets are more valuable samples for change detection, such as unlabeled ones, semantically ambiguous ones, pseudo-changes, and non-interesting changes, referred to as hard case samples (HCSs) in this article. There are two additional challenges to learning HCSs: 1) the loss function focusing on the foreground targets with rich labels and ignoring the HCSs in the background, called the imbalance problem and 2) it is difficult for a model to learn the change information of HCSs directly, which is called HCSs missingness. This article proposed a foreground association-driven Siamese network with hard sample optimization (HASNet). To deal with the imbalance problem, we propose an equilibrium optimization loss (EO-loss) function to regulate the optimization focus of the foreground and background, determine the HCSs through the distribution of the loss values, and introduce dynamic weights in the loss term to gradually shift the optimization focus of the loss from the foreground to the background hard cases as the training progresses. To address the HCSs missingness, we propose the scene-foreground association module by using potential remote sensing spatial scene information to model the association between the target of interest in the foreground and the related context to obtain scene embedding to reinforce the feature of hard cases. Experiments on four public datasets with 11 baselines show that HASNet outperforms current state-of-the-art CD methods, particularly in detecting HCSs. The source code is available at https://github.com/GeoX-Lab/HASNet.**

***Index Terms*— Change detection, foreground-scene association, hard case samples (HCSs), remote sensing image.**

I. Introduction

CHANGE detection is a crucial technology for the interpretation and analysis of remote sensing images. Remote sensing change detection (RS-CD) focuses on the same geographical areas on the Earth's surface to capture the changes in the land cover information over time and generate binary change maps representing differential information. To date, many CD algorithms and theoretical models have been proposed. Coupled with the development of remote sensing imaging technology, RS-CD has been widely applied in many significant fields, including land resource management [1], natural disaster monitoring [2], agricultural land analysis [3], and urban–rural development planning [4], [5].

In recent years, due to the gradual maturation of the commercial high-resolution remote sensing (HRRS) satellites, HRRS imagery has the ability to describe a variety of feature entities more comprehensively and in more detail [6], [7], [8]. However, the prevalence of light shading, target occlusion, and seasonal changes in optical remote sensing images, coupled with difficulties such as spectral feature confusion, increases the intraclass variance in the features, whereas the interclass variance decreases, which in turn leads to a significant reduction in the separability of the remotely sensed targets. This problem is particularly pronounced in HRRSs and, therefore, induces intractable CD problems involving hard samples, pseudo-changes, and non-interesting changes. Similar to semantic segmentation tasks, RS-CD relies heavily on the effectiveness of feature learning for key objects, which are also called foreground targets [9]. In addition to these foreground targets, the most valuable elements for model optimization in the image space are the semantically ambiguous samples in the background, that is, hard case samples (HCSs), which include hard samples, pseudo-changes, and non-interesting changes [10]. After many experimental explorations, we realized that HCSs are often more valuable for optimization in the later model training stages than foreground targets because, for the pixel-level prediction task, the deep neural network first obtains the foreground target from the background information and then calculates the conditional probability of the category to which the foreground target belongs on a pixel-by-pixel basis and selects the highest probability score as the category information for that pixel. The probability of an HCS being correctly discriminated can increase if and when the hard case target and the foreground target are optimized to the same degree. By contrast, HCSs interfering with the foreground target feature extraction and feature discrimination processes will have a negative impact on RS-CD, so optimization of the feature learning process for background hard cases will certainly be highly important.

Currently, deep learning techniques are increasingly widely used in remote sensing, and deep neural network-based change

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detection models are also focused on solving HCS problems and sub-problems. These methods can be broadly categorized into two groups following the approaches of 1) strengthening the feature representations of key information and 2) adding an attention mechanism and its variants. To enhance feature representation, Zheng et al. [11] considered the importance of the accuracy and completeness of building boundary recognition for CD and proposed a target edge guidance module and a feature differential enhancement module based on the transformer for refining edge features and fusing different levels of change information, respectively, thus enhancing the high-frequency information of buildings. Xu et al. [12] realized the impact of the structural information of remote sensing targets on the detection accuracy and proposed a multiscale context aggregation network in which the global attention pyramid module and the dense feature fusion module are used to enhance the depth features of the original target and bridge the semantic gap between multiscale features, respectively. For the approach of incorporating attention mechanisms, Fang et al. [13] introduced an integrated channel attention module for the deep supervision of the features of interest and designed a tightly coupled information transmission mechanism between encoders and decoders. This approach mitigates the loss of deep localization information in neural networks and ultimately enables the localization of edge pixels and the capture of small targets. Chen et al. [14] considered the prevalent pseudo-change problem in CD and solved the pseudo-change problem to a certain extent by applying spatial and channel attention mechanisms to sensitize the network to the changes of interest and additionally balancing the variability between the samples by penalizing non-interesting changes and increasing the attention given to the changes of interest. In the first category of methods, Chen et al. [15] used differential features at different scales to strengthen the feature representation of the changes, effectively mitigating issues such as pseudo-changes. However, the greatest challenge in optical remote sensing image feature extraction lies in learning accurate semantic features from HCSs, and strengthening only the feature representation and designing some kind of attentional mechanism cannot completely guarantee the accuracy and completeness of the semantic understanding of HCSs; thus, there are currently no learning methods available for HCSs. However, networks such as DMINet [16] and DASNet [14] have addressed several challenges, such as pseudo-changes and differences between the foreground and background samples, and have optimized the focus of the network via joint attention and spatial and channel attention mechanisms, thus gaining some robustness against pseudo-change issues. However, changing the network's attention to various types of samples does not improve the model's ability to identify HCSs because the HCSs tend to be weak, dispersed, or occupy a small volume in the feature space; moreover, it is difficult for them to dominate the gradient optimization during the training process. Additionally, they will not be subjected to the same optimization effort as the foreground samples, and they cannot easily focus on the attention mechanism.

In the HRRS, HCSs in the background are often represented in the image space as light shadows, object occlusions, and small targets. We realize that each of these cases can be considered as some form of "camouflage" of a simple sample, for example, "occlusion" is the covering of a part of the target with a mask, for example, the tree in the red box in Fig. 1 occludes the main body of the road. "Shadows" are dark areas on the same side of a 3-D target, such as the shaded area around a high-rise building in the yellow box in Fig. 1, whereas "small targets" can be regarded as scaled-down versions of conventional targets, such as the discrete distribution of remote sensing targets in the orange box in Fig. 1. Based on the above consideration, learning HCSs by the network can start from "simple samples," and whenever the model obtains an excellent learning effect on simple samples, it gradually can begin learning background hard cases. Experiments have proven that this process optimization results in a better learning effect on HCSs; this is in line with the core idea of curriculum learning [17], [18], [19], which allows the model to first learn relatively simple samples so that it can form basic concepts and patterns and then gradually present the model with more challenging hard samples as the model continues to mature. A large body of work has verified that this strategy is beneficial for learning complex concepts via neural networks.



Fig. 1. Sample display of an HCS in a remote sensing image. HCSs such as target occlusions, shadows, and small targets are shown in red, yellow, and orange boxes, respectively.

After analyzing the aforementioned issues, we propose a Siamese foreground association-driven HCS optimization network for remote sensing image change detection. Using scene context to deepen the understanding of background hard cases is an excellent strategy; therefore, we propose the foreground-scene association module that uses potential remote sensing spatial scene information to model the relationship between the targets of interest in the foreground and the associated context and activate the output of the HCSs in the background to increase the difference between the foreground and the background and, conversely, to reduce the difference between the foreground and the hard cases of interest and to solve leakage and misreporting in CD. By contrast, background samples are less likely to dominate the gradient during training, but HCSs are more valuable for model optimization in the later stages of training; however, the number of background hard cases is much smaller than the number of simple samples. To address this problem, we propose an equilibrium optimization loss (EO-loss) function so that the network focuses on the foreground target in the pre-training stage and the background hard cases in the post-training stage for balanced

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optimization. HASNet achieves absolute $F1$ scores reaching 92.42%, 98.11%, 93.14%, and 82.84%, respectively, on four datasets, CDD [20], LEVIR-CD [21], Google-CD [22], and SYSU-CD [23], and yields improvements of 2.29%, 2.61%, 6.34%, and 1.77%, for an average increase of 3.26% compared with 11 advanced baseline methods, such as USSFC-Net [24]. The contributions are listed as follows.

1) To address the common issue of HCSs in HRRS, we propose a Siamese foreground association-driven HCS optimization network for remote sensing image change detection. To address the "imbalance" in optimizing HCSs, we introduce an EO-loss function. By analyzing the distribution of loss values, HCSs are identified, and dynamic weights are introduced in the loss term; this gradually shifts the focus of loss optimization from the foreground to background hard cases, ensuring the successful discovery of the HCSs of interest.
2) To better understand the semantics of HCSs, we propose a foreground-scene association module. It uses latent remote sensing spatial scene information to model the association between the foreground targets of interest and related context, obtaining scene embeddings. These are then applied to the feature discrimination of HCSs. Moreover, this approach reduces the feature difference between the samples of interest in background hard cases and the foreground, thereby suppressing issues such as missed detections in CD.

The rest of this article is organized as follows. Section II reviews related work. Section III describes our method. Section IV describes the design of the validation experiments, followed by the analysis and discussion. Section V concludes this article and proposes research directions for future work.

II. Related Works

In recent years, deep learning technology has enabled qualitative advances in remote sensing image change detection, with many DL-based CD algorithms and theoretical models continuously proposed. Below, we review two aspects of the related work: CD methods based on two mainstream deep networks (CNNs and transformers) and the current status and issues of the optimization methods for HCSs.

A. CD Methods Based on CNNs or Transformers

The introduction of convolutional neural networks (CNNs) has driven rapid developments in RS-CD research, with existing studies divided into two main stages: 1) the optimization stage of the network structure and 2) the application stage of attention mechanisms. In the network structure optimization stage, CNN-based methods have evolved toward depth and modularity, aiming to obtain more abstract feature representations and achieve plug-and-play functionality, which aids in understanding complex objects and scenes. For instance, Zhan et al. [25] first organized the CNN backbone within a Siamese framework to obtain feature information and subsequently determined changes by measuring the distance between the features and threshold segmentation. Daudt et al. [26] proposed three FCN-based CD methods: FC-EF, FC-Siam-diff, and FC-Siam-conc and discussed the impact of network structures and feature fusion methods on CD tasks through comparisons. Numerous FCN-based CD networks have been proposed; for example, Guo et al. [27] proposed a fully convolutional metric network that learns implicit metric criteria to measure the similarity of feature mappings, thereby learning discrimination patterns for change and no change; Zheng et al. [28] introduced an end-to-end U-shaped FCN change detection network called CLNet that combines features of different scales and multilevel context information to increase CD effectiveness; Peng et al. [29] proposed a difference-enhanced dense-attention CNN that enhances the accuracy of change information retrieval by introducing dense attention and the DE unit. In the application stage of attention mechanisms, numerous works guide the network to learn the features of the emphasized part through attention mechanisms, thus learning key discriminative information related to changes. For example, Chen et al. [14] introduced a dual-attention mechanism (spatial and channel attention) into a fully convolutional Siamese network, solving problems such as pseudo-changes by enhancing the focus on the changes of interest.

In recent years, there has been a surge of work addressing the complexity and semantic uncertainty of remote sensing targets by introducing context modeling to enhance the learning capability of neural networks, which has proven to be crucial for RS-CD tasks. Current context modeling methods include four mainstream methods: multiscale feature fusion, deep supervision network architecture, application of dilated convolution, and design of various attention mechanisms. Many works combine the advantages of several mainstream methods, integrating them into a single network. For example, Jiang et al. [30] used various methods to merge low-level and co-attention-level features and establish long-range contextual connections. Wang et al. [31] proposed a depth-supervised network based on self-attention that extracts multilevel image features during the encoding phase. To further highlight change features, they proposed an adaptive attention mechanism that combines spatial and channel features, capturing the relationships between the change features of different scales. While the use of CNNs to obtain multiscale feature mappings of targets is important for CD tasks, fundamentally, CNNs cannot model long-range dependencies. Fortunately, the advent of transformers has changed this situation.

Dosovitskiy et al. [32] first applied a transformer to image classification tasks in 2018, achieving remarkable results; moreover, the transformer also led to significant advances in the application of RS-CD. Chen et al. [33] were the first to apply transformers to CD tasks and proposed the bitemporal image transformer (BIT) network; its encoder, which is embedded with a transformer, models the spatiotemporal context of the abstract feature mappings obtained from CNNs. Then, tokens enriched with contextual information are fed back to the pixel space through the transformer decoder to locate the changes of interest. BIT once again proved the superiority of the transformer in context modeling. Since contextual information is crucial for semantically understanding remote sensing imagery, many RS-CD networks based on ViT have

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been proposed. Currently, transformer-based methods can be divided into two main categories: hybrid transformer methods and CNNs. For example, Feng et al. [34] reported that the sequential use of a CNN and ViT hindered the interaction of depth and breadth features; consequently, they proposed a parallel method to realize the coupling and complementarity of local and global features. Li et al. [35] found the advantages of U-Net and ViT to be complementary, and creatively embedded ViT into a U-shaped network, overcoming the difficulty of feature layer relationship modeling and the inaccuracy of differential feature representation. Xu et al. [36] adopted progressive sampling ViT to eliminate irrelevant change interference to address pseudo-changes and missed detections and subsequently applied a fusion module to obtain complete edge information, ensuring the accuracy of the change information. The other category includes pure transformer-based CD methods, such as those used by Yan et al. [37], who introduced a pyramid structure to aggregate multilevel features of the transformer to solve irregular change area boundaries via a progressive attention module and increase the representation level of interdependent features. Similarly, Zhang et al. [38] designed a dual U-shaped CD network using the Swin-T backbone, aiming to completely overcome the inherent limitations of convolutional operations and fully exploit the global modeling advantage of the transformer.

B. Current Research Status of Optimization Methods for HCSs

HCS problems are widespread in machine learning and data science, and typically refer to situations where certain samples are more difficult than others to classify, learn, and predict correctly during model training. Several aspects of the HCS problem in remote sensing imagery can be summarized as follows: 1) sample imbalance; 2) small targets; 3) shadow coverage and object occlusion; and 4) pseudo-changes. Many studies have explored the above problems in computer vision tasks [39]; for example, Zheng et al. [40] proposed a foreground-aware relation network (FarSeg) from the perspectives of relation-based and optimization-based foreground modeling, alleviated the problem that high spatial resolution remote sensing images always face large-scale variation, larger intra-class variance of background, and foreground-background imbalance, which affect the model segmentation performance. Li et al. [41] proposed a point-wise affinity propagation module based on the FPN framework, named PointFlow, which alleviated the problem that recent dense affinity context modeling performs poorly compared to baselines due to over-introduced background context. For the sample imbalance problem, the current mainstream approach is to optimize the loss function design. Lin et al. [42] proposed focal loss for object detection tasks, introducing an adjustable focus parameter to reduce the weight of the easy-to-classify samples and focusing more on the HCSs, whereas weighted cross-entropy loss [43] adjusts the model's focus on different categories by introducing different weights for each class, thus solving the sample imbalance problem. Zhao et al. [44] introduced Dice loss in medical image segmentation tasks, minimizing the Dice coefficient to increase the sensitivity of the model to situations with fewer pixels and suppress the background weight; however, Dice loss can be used only for binary segmentation tasks. To address issues such as small targets and pseudo-changes, in addition to data augmentation operations, optimization of the model structure is vital. For instance, Lin et al. [45] proposed the FPN method, which generates feature maps of different resolutions at different levels of the network. High-level features contain stronger semantics, and low-level features contain more details, enhancing the perception of small targets. Fu et al. [46] introduced a dual-attention mechanism in semantic segmentation tasks that use spatial attention to capture key spatial information and use channel attention to capture the features more relevant to the task, thereby improving key information extraction. Research on CD has also actively explored optimization solutions for HCSs. For example, MFCN [47] designs multiscale convolutional kernels to extract detailed surface features and combines the WBCE and Dice loss for balanced sample optimization, improving the model's ability to discriminate minute features from two aspects. Zhu et al. [48] also focused on the problems such as small target changes and edge pixel misclassification; proposed ECFNet; and designed three processes of feature extraction, feature comparison, and feature fusion. In the fusion process, the number of channels is constrained to better utilize the fine-grained information in multiscale features for result prediction. Guo et al. [49] realized the importance of interest feature extraction and feature fusion for CD and proposed three modules, namely deep multiscale feature extraction, parallel convolutional feature fusion, and self-attention-based feature refinement, to integrate multiscale change information and further enhance the feature representation of interest targets. The current optimization methods for HCSs focus mainly on model structure improvement, interaction of deep and shallow layer information, and optimization of attention mechanisms. However, the idea of this article is to start from the learning rules of HCSs and divide the feature learning process according to difficulty characteristics. Sample optimization can be achieved through two steps: HCS mining and balanced optimization of foreground and hard cases. In this process, background hard cases are enhanced using relevant scene information in the foreground, thus improving feature discriminability and accuracy. Therefore, we design a foreground-scene association module and an EO-loss to learn the HCSs commonly found in the background.

III. Methodology

In this section, we introduce the proposed CD method in detail. Section III-A presents the overall framework of HASNet. Section III-B introduces the variant feature pyramid network (V-FPN) encoder. Section III-C describes the foreground-scene association module. Section III-D covers the dual-temporal feature fusion and feature decoding structure, and EO-loss.

A. Network Architecture

As shown in Fig. 2, the overall structure of HASNet is an end-to-end Siamese network. A pair of dual-temporal

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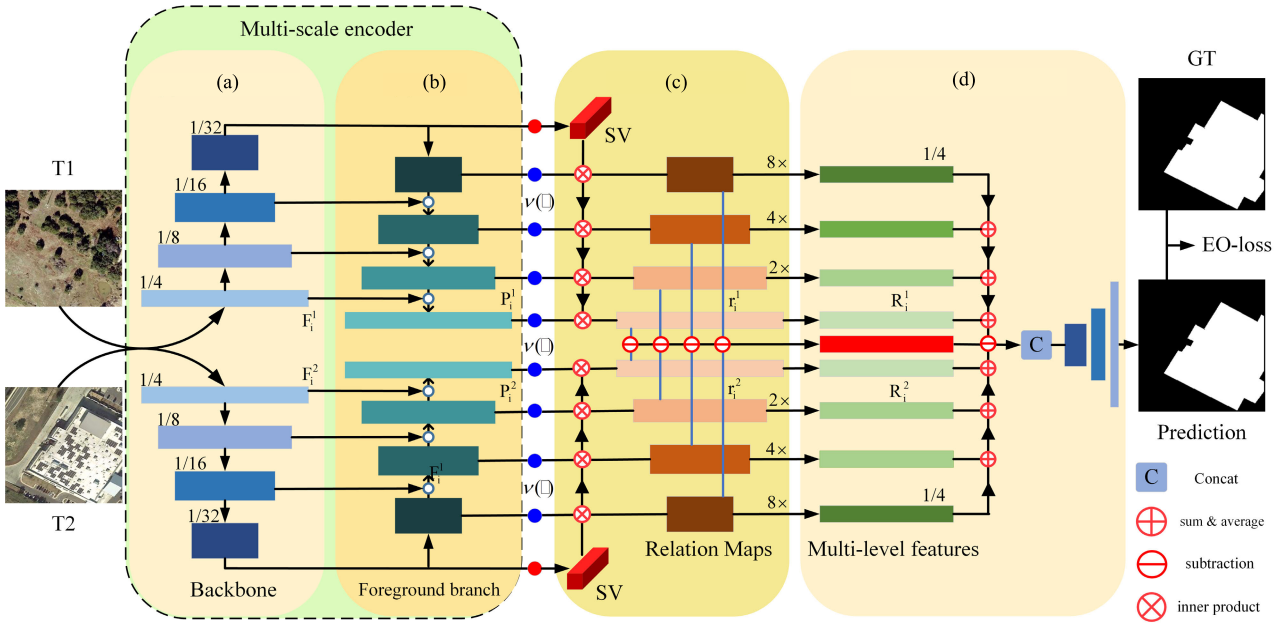


Fig. 2. Overview of HASNet. (a) PVT/ResNet. (b) FPN. (c) FS-relation. (d) Fusion and decoder.

remote sensing images is taken as input, and a binary change map is output. The network consists of a V-FPN module, a foreground-scene association module (FS-relation), an EO-loss, and a feature fusion and lightweight decoder. The V-FPN module is responsible for extracting multiscale target feature and obtaining deep feature information. The FS-relation module is built on a collection of multiscale feature maps. Our idea is that change misses and false detections are due to numerous difficult-to-distinguish samples in the background, whereas the foreground lacks sufficient discriminative information for interest targets. Therefore, the FS-relation module is designed to improve the associative capability of interest features, thereby enhancing the discrimination ability for hard case targets. Additionally, to more effectively learn about hard case targets in the background and interest targets in the foreground, we propose an EO-loss for balanced optimization and hard case information learning of foreground–background samples. Finally, we design a multilevel feature fusion module and a lightweight decoder for fusing dual-temporal change information and restoring it to the original pixel space to obtain the final change map.

B. Foreground-Scene Association Module

1) V-FPN Encoder: At present, many works have focused on the modeling of multiscale FPN to extract rich multiscale features [50], [51], [52]. To obtain deep features of scene embedding and utilize scene embedding at multiple scales to enhance feature representation, we also made some changes in FPN and designed the V-FPN for multiscale feature extraction and adaptation to multilayer scene information embedding. The V-FPN consists of a feature extraction branch and a scene information embedding branch. Both branches utilize a variant FPN network divided into multiscale feature layers. The FPN, originally proposed by Liu et al. [6] for object detection tasks, aims to capture strong semantic information in multiple feature layers. To further enhance the model's multiscale feature extraction and fusion capabilities, we design each layer of the FPN with ResNet as the backbone network for basic feature information extraction. As shown in Fig. 2, given a pair of input images $T^m, m = 1, 2$, the size of the feature layers in the feature extraction backbone network decreases vertically; this results in a multilevel feature layer set $F_i^m, i = 1, 2, 3, 4; m = 1, 2$, with the output strides relative to the input image of (4, 8, 16, 32) pixels. Building on the standard FPN structure, we use feature layers from deep to shallow and lateral feature connections to generate a pyramid feature mapping set $P_i^m, i = 1, 2, 3, 4; m = 1, 2$ with the same dimensions. This method fully links spatial detail information of shallow features with strong semantic information of deep features, aiding in the modeling of multiscale target context information. This process can be expressed by the following equation:

$$P_i^m = \tau\left(F_i^m\right) + \sigma\left(P_{i+1}^m\right), \quad i = 1, 2, 3, 4; \ m = 1, 2 \tag{1}$$

where τ represents a learnable 1×1 convolutional layer used for lateral feature connections and where σ represents a nearest-neighbor upsampling operation with a dimension of 2.

In the final layer F_4 of the pyramid feature set, we design an important feature flow branch, namely, the foreground-scene information embedding branch. This structure is based on global context aggregation to obtain the spatial scene embedding vector SV, which models the dependency relationship between geographical scenes and interest targets, including interest targets in the foreground and HCSs. The specific structure and the principle of this process are introduced in Section IV. Additionally, in our experiments, we used two types of backbone networks for feature extraction: ResNet [53] and the pyramid vision transformer [54]. The former, with its ingenious residual modules, has strong feature extraction capabilities. Considering accuracy and efficiency, we chose ResNet50 as the first backbone network. Moreover, the PVT can consider both low-level local features and high-level global features. The transformer can learn long-range dependencies

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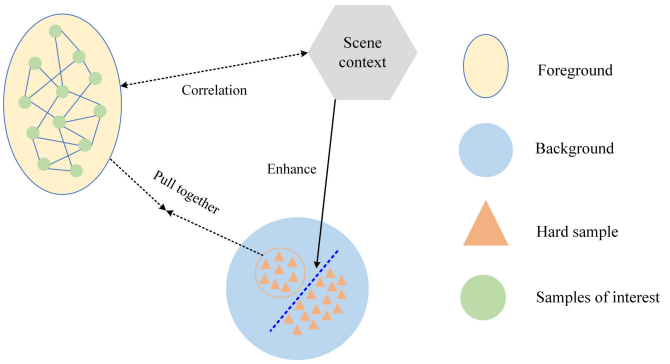


Fig. 3. Schematic of the FS-relation module.

between features, endowing it with strong feature representation capabilities at different scales. In the experiments, we used PVTv2 as the second backbone network.

2) Foreground-Scene-Associated Information Embedding: HRRS is characterized by rich spectral features and complex scene information. Issues such as varying lighting conditions, seasonal changes, and shadows and occlusions caused by tall buildings commonly occur; this means that there is a significant difference in the image spectral features and a large intraclass variance in the background, specifically manifested as a large number of HCSs in the background information, leading to problems such as false positives, false negatives, and pseudo-changes in CD. To address this issue, we propose a foreground-science association module that leverages the scene context of the remote sensing space to model the feature representation of the foreground targets positively and to enhance the feature discrimination ability of background samples. The main principle is illustrated in Fig. 3. The FS-relation module first explicitly models the foreground targets and geographic scene information and then uses implicit geographic scenes to associate the foreground with the relevant context. This relationship is subsequently used to enhance the input feature mapping, increasing the difference between the foreground targets and background hard cases and reducing the likelihood of false alarms and misreporting.

As depicted in Fig. 4, the FS-relation module generates a new feature mapping set R_i^m based on the feature pyramid collection $P_i^m, i = 1, 2, 3, 4; m = 1, 2$. This module initially re-encodes each layer of P_i^m to preliminarily form the feature mapping Q_i^m. This mapping is then reweighted according to the correlation map r_i^m, ultimately producing the relation-enhanced feature mapping set R_i^m. The correlation map r_i^m refers to the similarity matrix between the geographical scene with scene graph relationship and the foreground representation [55], [56]. To achieve more discerning R_i^m, the model learns a feature mapping function to align the feature input P_i^m with the scene vector (SV) into the same dimension, facilitating feature interaction. Here, $Q_i^m \in R^{d\times H\times W}$ is transformed from the feature pyramid layer $P_i^m \in R^{d\times H\times W}$ through the scale projection function $v(\cdot)$, as expressed in the following equations:

$$v(\bullet) : R^{C\times H\times W} \rightarrow R^{d\times H\times W} \tag{2}$$

$$Q_i^m = v\left(P_i^m\right). \tag{3}$$

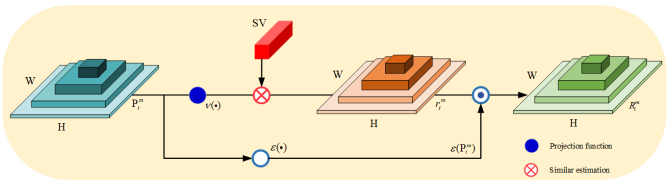


Fig. 4. Schematic of scene information embedding.

We implement $v(\bullet)$ efficiently through a 1×1 convolution layer followed by batch normalization (BN) and ReLU.

The similarity relation matrix set r_i^m is due to the interaction between the scene embedding information and the foreground pyramid features. Therefore, to compute this set, the 1-D scene embedding information SV $\in R^d$ interacts with the foreground feature projection Q_i^m. Here, the scene embedding information SV is obtained by linearly projecting P_4^m, as represented by the following equation:

$$SV = \varphi\left(P_4^m\right). \tag{4}$$

$\varphi(\bullet)$ signifies a scene-focused projection function. For ease of computation, this process is executed by a learnable 1×1 convolution, with the output channel number set to d. For the same pair of input images, the remote sensing scene information should be identical, allowing each layer of features in Q_i^m to share an SV. At this stage, the similarity relation matrix set r_i^m is derived using the following equation:

$$r_i^m = \delta\left(SV, Q_i^m\right) = SV \odot Q_i^m \tag{5}$$

where $\delta(\bullet)$ denotes the similarity measure function. To simplify the computation and improve the efficiency, the similarity measure is implemented by the vector inner product.

The details of the foreground-scene correlation process are illustrated in Fig. 4. The final relation-enhanced feature mapping R_i^m is computed according to the following equation (6):

$$R_i^m = \frac{\varepsilon\left(P_i^m\right)}{1 + \exp\left(-r_i^m\right)}. \tag{6}$$

$\varepsilon(\bullet)$ is a feature re-encoding structure designed to re-encode the pyramid feature mapping set P_i^m, similar to the aforementioned $v(\bullet)$ structure. In this equation, we use a polynomial containing r_i^m to weight and highlight the re-encoded feature mapping $\varepsilon(P_i^m)$.

C. Lightweight Decoder and Dual-Branch Information Fusion

As shown in Fig. 1, to capture multiscale change information, we fuse the relationship-enhanced feature mapping sets P_i^1 and P_i^2 of the dual branches, which means that the multiscale feature map C_i can be derived according to the following equation:

$$C_i = abs\left(R_i^1 - R_i^2\right). \tag{7}$$

To restore the spatial resolution of the relationship-enhanced feature mapping set, we design a lightweight decoder; the structure of which is shown in Fig. 5. According to the foreground-scene correlation module, given the

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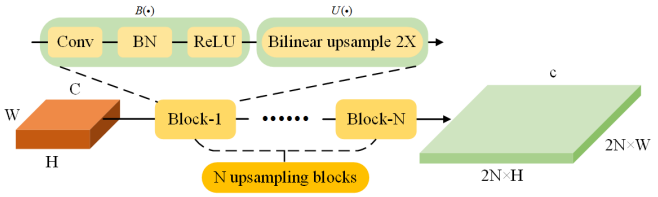


Fig. 5. Lightweight decoder architecture diagram.

relationship-enhanced feature mapping R_i^m, the upscaled feature mapping set $S_i^m \in R^{c\times\alpha H\times\alpha W}$ can be computed by this decoder. This decoder consists of several upsampling modules, each containing two parts: a dimension transformation operation $B(\bullet)$ and a bilinear interpolation upsampling operation $U(\bullet)$. These are responsible for transforming multiscale feature channels and restoring feature dimensions, respectively, as represented by the following equation:

$$\hat{S}_i^m = U_N\left(B_N\left(S_i^m\right)\right). \tag{8}$$

Here, N represents the number of times that the upsampling module is used, which is determined by the dimensions of the R_i^m feature layer. The $B(\bullet)$ operation is achieved through a 3×3 convolution, BN, or ReLU operation, whereas $U(\bullet)$ represents bilinear interpolation upsampling with a scaling factor of 2; thus, the upsampling scale is $2N$. Each layer of the upscaled feature mapping set $\hat{S}_i^m$ is subsequently aggregated from high to low to obtain a semantically rich feature mapping $M^m, m = 1, 2$; this is achieved through pointwise mean computation for feature fusion, followed by a 1×1 convolution for feature integration and parameter computation, as described by the following equation:

$$M^m = \mathrm{Conv}_{1\times1}\left(\frac{\sum_{i=1}^{4}\hat{S}_i^m}{\max(i)}\right), \quad m = 1, 2. \tag{9}$$

For ease of expression and clarity, we collectively refer to the upsampling and aggregation modules as deo(•). Then, the change feature t can be further fused from the dual-branch aggregated features M^m, as described by the following equation:

$$t = abs\left(M^1 - M^2\right). \tag{10}$$

Finally, to better represent the multiscale information in the final predictive map, a skip connection is established. The specific operation involves aggregating the change feature t with the multiscale feature information C_i, as represented by the following equation:

$$T = \mathrm{Concat}(t, \mathrm{deo}(C_i)). \tag{11}$$

We successfully obtained the semantically enriched variational feature T. To derive the final predictive change map T_{change}, it is imperative to remap the advanced semantic-laden features T back into the original pixel space. To address this issue, we craft a lightweight deconvolution decoder based on the fully convolutional network (FCN) framework. This architecture is segmented into three core components: the feature fusion layer, the deconvolution layer, and the classification layer. Each segment is intricately designed, comprising a 3×3 convolution layer with BN and ReLU, a 4×4 deconvolution layer also equipped with BN and ReLU, and, finally, a deconvolution layer equipped solely with BN. The culmination of this process is the pixel-level classification to discern change information, executed through the sigmoid function, as described by the following equation:

$$T_{\mathrm{change}} = \mathrm{Sigmoid}(\eta(\mathrm{Conv}_{3\times3}(T))). \tag{12}$$

Here, $\eta(\bullet)$ symbolizes a sequence of two consecutive deconvolution operations.

D. Equilibrium Optimization Loss Function

Remote sensing imagery backgrounds usually contain a certain number of HCSs, and models often show poor learning effectiveness for these samples. One reason for the poor learning effectiveness is that the number of HCSs is relatively small, and only a fraction of the HCSs related to interest areas are meaningful for model optimization in the later stages of training. Another reason is that, in the early stages of training, the model's judgment of HCSs is uncertain, and direct learning from these samples is less effective. Therefore, we designed the EO-loss to address these issues. The training loss measures the feature distance between the target supervision and the predicted values. The difference between the model's predicted values and the true values for HCSs is often greater than that for simple samples. Thus, the distribution of HCSs can be roughly estimated by the loss values. The loss value is positively correlated with the difficulty of sample learning and is generally expressed as follows: the more difficult the sample is to learn, the greater the loss value; the easier the sample is to learn, the smaller the loss value. In the later training stages, the model's predictions for simple samples have already reached good performance, and at this time, only the more challenging samples in the background and foreground are highly significant for training optimization. On this basis, we propose the EO-loss. As shown in Fig. 6, this loss function consists of three steps: HCS assessment, dynamic weight optimization, and backpropagation.

To obtain weights that represent the difficulty level of sample data, thereby adjusting the loss distribution pixel by pixel to achieve balanced optimization, we are inspired by the focal loss [42] approach for optimizing hard samples. We use an exponential polynomial with trainable parameters to predict the distribution of HCSs, which represents the predicted probability and the focus factor. For the pixel-level prediction task of distinguishing between the foreground targets and background HCSs, we aim to adjust the distribution of the loss without changing its total value to avoid the vanishing gradient problem. Therefore, we introduce a normalization parameter Z to eliminate the impact of outliers and anomalies on the overall results, ensuring stable and efficient learning of HCSs. This parameter Z must ensure that (13) is valid.

$$\sum_{i=0}^{H\times W}\mathrm{loss}(p_i, y_i) = \frac{1}{Z}\sum_{i=0}^{H\times W}(1-p_i)^{\gamma}\,\mathrm{loss}(p_i, y_i). \tag{13}$$

Here, $\mathrm{loss}(p_i, y_i)$ represents the binary cross-entropy loss value for the ith pixel, which can be calculated from the predicted probability value p_i and the actual value y_i. Therefore, the weight of each pixel's loss value is $(1/Z)(1-p)^{\gamma}$.

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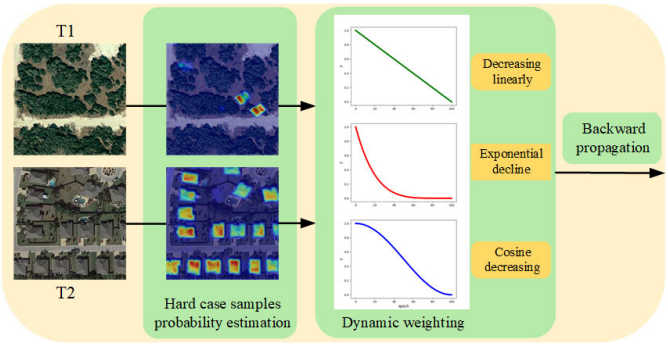


Fig. 6. EO-loss function structure.

TABLE I
INFORMATION FOR DIFFERENT WEIGHTING FUNCTIONS

Function	Expression	hyperparameter
Linear	$\lambda(t) = 1 - \frac{t}{\text{step}}$	step
Exponential	$\lambda(t) = \left(1 - \frac{t}{\text{step}}\right)^{\text{decaty}}$	step and decay
Cosine	$\lambda(t) = \frac{1}{2}\left(1 + \cos\left(\frac{t\pi}{step}\right)\right)$	step

The learning effectiveness of HCSs depends on the optimization method of the model. In the early stages of training, the model's confidence in sample category determination is not high, and dealing with too many HCSs at this time may lead to an unstable learning process. To ensure stable and convergent model training, we propose a dynamic weighting strategy based on a decay function. This strategy focuses on learning from simple samples in the early training stages and strengthens the optimization of HCSs in the later stages. Based on the binary cross-entropy loss BCE(•), we can express the EO-loss as shown in the following equation:

$$\text{loss} = \left[\lambda(t)\left(1 - \frac{1}{Z}(1 - p_i)^{\gamma}\right) + \frac{1}{Z}(1 - p_i)^{\gamma}\right]\text{BCE}(p_i, y_i). \tag{14}$$

Here, $\lambda(t)$ represents a weight function related to the training process t, which is used for weighting non-HCSs, and $\lambda(t) \in (0, 1)$ decreases monotonically. Therefore, as training progresses, the model's confidence in predicting non-HCSs increases continuously, and the focus of the loss value distribution gradually shifts to HCSs. Considering that there are multiple methods to decrease $\lambda(t)$, we design three representative and computationally convenient weight functions for weighting non-HCSs, with specific details shown in Table I. A linear decrease is used for the model to transition smoothly from simple to HCSs; an exponential decrease is based on the idea that the learning process of simple samples progresses very quickly, hence further shifting the optimization focus toward HCSs and increasing the time for optimizing the HCSs; and a cosine decrease distributes the optimization weights evenly between simple and HCSs, reducing the proportion of the transition process.

IV. EXPERIMENTS AND DISCUSSION

To evaluate the performance of HASNet, we conducted experiments on four public CD datasets, LEVIR-CD, Google-CD, CDD, and SYSU-CD, and compared the results with those obtained by the most recently reported RS-CD methods. Additionally, we designed ablation studies and effectiveness experiments and examined the experimental results.

A. Datasets

1) LEVIR-CD [17]: The large-scale building change detection dataset consists of 637 pairs of Google Earth images; each with a size of 1024 × 1024 pixels and a resolution of 0.5 m, spanning from the time period from 2002 to 2018; it contains a variety of building change information, such as villas, large factories, high-rise apartments, and garages.

2) Google-CD [18]: This dataset records changes in the various types of buildings, including large factories, villages, and warehouses, in the suburbs of Guangzhou, China, from 2006 to 2019. The 19 pairs of images are sourced from Google Earth, with spatial resolutions of 0.55 m and image sizes ranging from 1006 × 1168 to 493 × 5224.

3) CDD [16]: This dataset contains 11 pairs of multisource remote sensing images collected in different seasons; 7 pairs have a resolution of 7425 × 2202 pixels, and 4 pairs have 1900 × 1000 pixels, with resolutions varying from 0.03 to 1 m. The challenge of this dataset lies in accurately detecting the changes in buildings, roads, farmland, and vehicles, regardless of the impact of seasonal changes.

4) SYSU-CD [19]: This dataset includes 20 000 pairs of aerial images collected in Hong Kong, China, from 2007 to 2014; all of the images have a size of 256 × 256 pixels and a resolution of 0.5 m. The main types of changes in the data include urban buildings, road construction, marine construction, and vegetation changes.

B. Metrics

1) Evaluation Criteria: To evaluate the performance of the model, we used seven evaluation metrics to present the results: precision (P), recall (R), $F1$ score ($F1$), intersection over union (IOU), mean IOU (mIOU), overall accuracy (OA), and the kappa coefficient (Kappa). In the RS-CD tasks, higher precision indicates more correct detections in positive cases, higher recall means fewer losses in predicted positive cases, and larger $F1$, IOU, and mIOU values indicate better CD performance. A higher kappa coefficient indicates better consistency between two sets of predictions, whereas OA provides an overall assessment of all predictions that are classified correctly. The calculation formulas for these metrics are as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$F1 = \frac{1}{P^{-1} + R^{-1}} \tag{17}$$

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{18}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \tag{19}$$

$$\text{mIOU} = \frac{1}{2}\left(\frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} + \frac{\text{TN}}{TN + \text{FN} + \text{FP}}\right) \tag{20}$$

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$$\text{Kappa} = \frac{\text{OA} - \text{Pe}}{1 - P}. \tag{21}$$

2) Implementation Details: We conducted experiments using an NVIDIA 4080Ti graphics card and trained the model with a mini-batch Adam optimizer. The PyTorch deep learning framework was used. The number of epochs was set to 150, the batch size was set to 16, the learning rate was set to 5×10^{-4}, the weight decay was set to 5×10^{-4}, the step size was set to 50, and the momentum was set to 0.9. For the CDD dataset, we cropped the original images into small images of 256×256 pixels and randomly performed simple operations such as rotation, flipping, and central cropping. We obtained 10 000, 3000, and 3000 pairs of image patches for training, validation, and testing, respectively. For the LEVIR-CD dataset, we cropped the original images into non-overlapping blocks of 256×256 pixels, similarly performing random rotations, flipping, and central cropping for data augmentation. We divided the images randomly into three sets with 8392/600/1200 pairs used for training, validation, and testing. For the Google-CD dataset, we cropped the original images into blocks of 256×256 pixels and performed random rotations, flipping, and central cropping steps; we obtained 2661/157/312 pairs of images for training, validation, and testing. For the SYSU-CD dataset, we cropped the original images into small images of 256×256 pixels, similarly performing random rotations, flipping, and central cropping, and obtained 14 001/1999/4000 pairs of images for training, validation, and testing.

C. Results and Discussion

1) Benchmark Methods: In this section, we introduce the benchmark methods used for comparison. To demonstrate the superiority of HASNet, we selected the classic research achievements for each subdirection for comparison, including three pure convolution-based CD methods: FC-EF, FC-Siam-Di, and FC-Siam-Co; the DASNet method based on channel and spatial attention; the BiT method embedding ViT into the CNN backbone; the SwinSUNet pure Swin-T-based method; DMINet, which is a method based on cross-temporal joint attention; the ICIF-Net method based on cross-interaction attention and multiscale feature fusion; the EGCTNet CD method for building edge feature enhancement; the HFA-Net method based on high-frequency information enhancement and spatial attention; and the USSFC-Net 3-D attention method, which is based on spatial–spectral feature synergy.

1) *FC-EF [26]:* An early feature fusion CD method. It directly concatenates dual-temporal images in the channel dimension before feeding them into the network.
2) *FC-Siam-Co [26]:* A weight-sharing dual-encoder Siamese network that acquires change information by fusing features at various levels.
3) *FC-Siam-Di [26]:* This method differs from FC-Siam-Co in that it changes the feature map fusion method in the network from concatenation to absolute difference.
4) *DASNet [14]:* Based on VGG16 or ResNet50, this Siamese network introduces spatial and channel attention in the feature encoding phase to enhance the network's focus on interest changes and resist pseudo-change interference. The change map is obtained using a feature metric module.
5) *BIT [33]:* This method embeds a transformer module in the convolutional network, converting dual-temporal feature mappings into compact semantic tokens to facilitate the modeling of the spatiotemporal context in the feature space, thereby optimizing feature representation in the image space.
6) *SwinSUNet [38]:* Based on Swin-T, this dual U-shaped CD network uses multiple pure transformer encoders to capture the spatiotemporal context of dual-temporal images. After feature fusion and linear mapping, the change information is decoded using multiple Swin-T blocks combined with skip connections.
7) *DMINet [16]:* Based on dual-branch ResNet18 for feature extraction, DMINet unifies self-attention and cross-attention in one module to guide the global feature distribution of each input. Two change information acquisition structures are also designed, namely, the fusion method, which is based on subtraction and concatenation, and a multilevel differential aggregation method, which is based on incremental feature alignment.
8) *ICIFNet [34]:* This method designs a four-branch cross-interaction feature extraction structure with parallel CNNs and ViT to promote the interpenetration of local and global features. Mask-based aggregation and spatial alignment (SA) schemes are also introduced for scale integration, achieving information integration at different resolutions.
9) *EGCTNet [57]:* Focusing on the issue that local fine features and global information in CD cannot be obtained simultaneously, a fusion encoder is designed that combines a CNN and transformers. Additionally, an edge detection branch is proposed that uses edge information to guide mask feature generation.
10) *HFA-Net [11]:* To address the issue of insufficient high-frequency information acquisition for clearly defined targets, a high-frequency attention-guided module is proposed. It consists of two main stages: first, spatial attention is used to search for and focus on buildings; second, high-frequency enhancement is used to highlight the high-frequency information of features, which better represents the edges of changed buildings.
11) *USSFC-Net [24]:* Simultaneously modeling spatial and spectral features, this method proposes an efficient collaborative network to generate 3-D attention for richer feature information. Additionally, a multiscale decoupled convolution is designed to flexibly capture the multiscale features of changing objects.

2) Performance Comparison: To assess the effectiveness of HASNet accurately, we conducted quantitative experiments on four public CD datasets. Tables II–V show the accuracies of HASNet on the LEVIR-CD, Google-CD, CDD, and SYSU-CD test sets, respectively. The quantitative results indicate that our method consistently outperforms the other methods on three datasets and demonstrates significant advantages on the SYSU-CD dataset. For example, on the first three datasets, HASNet

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TABLE II
COMPARISON RESULT ON LEVIR-CD TEST SET

Method	Year	Precision	Recall	F1	OA	mIOU	IOU	Kappa
FC-EF	2018	86.91	80.17	83.40	98.39	80.98	71.53	80.44
FC-Siam-Di	2018	89.53	83.31	86.31	98.67	83.87	75.92	84.28
FC-Siam-Co	2018	91.99	76.77	83.69	98.49	81.42	71.96	80.94
DASNet	2020	80.76	90.70	79.91	94.32	79.22	74.65	75.39
BIT	2021	89.24	89.37	89.31	98.92	89.02	80.68	88.35
SwinSUNet	2022	88.03	84.76	86.37	99.16	87.57	76.01	85.93
DMINet	2022	**93.02**	89.58	91.26	99.46	91.69	83.94	90.99
ICIFNet	2022	92.61	90.80	91.69	99.48	92.07	84.66	91.43
EGCTNet	2022	87.66	91.42	89.50	99.33	90.15	81.00	89.15
HFA-Net	2022	90.10	80.48	85.02	99.11	86.51	73.94	94.56
USSFC-Net	2023	91.06	88.72	89.81	99.37	90.48	81.61	89.55
Ours	-	92.80	**92.04**	**92.42**	**99.53**	**92.71**	**85.90**	**92.17**

TABLE III
COMPARISON RESULT ON GOOGLE-CD TEST SET

Method	Year	Precision	Recall	F1	OA	mIOU	IOU	Kappa
FC-EF	2018	80.81	64.39	71.67	85.85	67.83	55.85	66.40
FC-Siam-Di	2018	85.44	63.28	72.71	87.27	69.37	57.12	70.01
FC-Siam-Co	2018	82.07	64.73	72.38	84.56	68.39	56.71	69.05
DASNet	2020	69.92	68.83	77.46	94.59	79.22	74.65	75.39
BIT	2021	92.04	72.03	80.82	96.59	79.11	67.81	77.33
SwinSUNet	2022	90.77	75.92	82.68	96.59	83.39	70.48	80.81
DMINet	2022	93.06	90.92	91.98	98.30	91.63	85.15	91.03
ICIFNet	2022	92.37	91.02	91.69	98.23	91.35	84.66	90.70
EGCTNet	2022	90.34	86.23	88.24	97.53	88.12	78.95	86.86
HFA-Net	2022	81.97	77.91	79.89	95.80	80.97	66.52	77.55
USSFC-Net	2023	90.58	83.33	86.80	97.29	86.86	76.69	85.3
Ours	-	**95.13**	**91.25**	**93.14**	**98.56**	**92.79**	**87.17**	**92.34**

TABLE IV
COMPARISON RESULT ON CDD TEST SET

Method	Year	Precision	Recall	F1	OA	mIOU	IOU	Kappa
FC-EF	2018	65.60	55.01	57.65	93.58	65.32	52.20	52.42
FC-Siam-Di	2018	78.25	65.76	69.04	94.44	67.24	56.57	62.39
FC-Siam-Co	2018	74.51	73.87	71.16	94.92	69.48	60.11	66.44
DASNet	2020	90.00	90.50	91.00	99.10	90.28	83.01	90.12
BIT	2021	96.19	93.99	95.07	98.62	94.43	90.61	93.00
SwinSUNet	2022	95.70	92.30	94.00	98.50	93.36	91.74	93.28
DMINet	2022	98.40	94.66	96.49	99.15	96.13	93.23	96.01
ICIFNet	2022	**98.81**	94.35	96.53	99.16	96.17	93.29	96.05
EGCTNet	2022	94.32	86.74	90.37	97.72	89.94	82.43	89.08
HFA-Net	2022	92.23	62.55	74.54	94.73	76.85	59.42	71.72
USSFC-Net	2023	96.19	95.46	95.92	98.97	95.41	91.99	95.24
Ours	-	98.32	**98.12**	**98.21**	**99.56**	**98.00**	**96.49**	**98.33**

TABLE V
COMPARISON RESULT ON SYSU-CD TEST SET

Method	Year	Precision	Recall	F1	OA	mIOU	IOU	Kappa
FC-EF	2018	74.32	75.84	75.07	86.02	73.62	60.09	69.37
FC-Siam-Di	2018	89.13	61.21	72.57	82.11	68.77	56.96	67.01
FC-Siam-Co	2018	82.54	71.03	76.35	86.17	74.21	61.75	70.09
DASNet	2020	68.14	70.01	69.14	80.16	56.88	60.65	64.37
BIT	2021	82.18	74.49	78.15	90.18	75.29	64.13	70.93
SwinSUNet	2022	78.27	73.93	76.04	89.01	74.02	61.34	68.92
DMINet	2022	82.08	**84.86**	**83.45**	92.06	80.53	**71.60**	78.23
ICIFNet	2022	78.64	77.75	78.20	89.77	75.84	64.21	71.51
EGCTNet	2022	75.92	79.81	77.82	89.27	75.24	63.69	70.75
HFA-Net	2022	80.93	70.91	75.60	89.20	73.90	60.76	68.70
USSFC-Net	2023	79.84	82.34	81.07	90.93	78.45	68.16	75.11
Ours	-	**87.21**	78.88	82.84	**92.29**	**80.62**	70.70	**78.28**

outperforms the other compared methods in most accuracy metrics, with $F1$ scores exceeding those of the latest method, DMINet, by 1.16%, 1.15%, and 1.72%, respectively. Our method achieves certain breakthroughs in all seven accuracy metrics on all four datasets, confirming the importance of introducing an EO-loss function and a novel CD network architecture with a foreground-science association module.

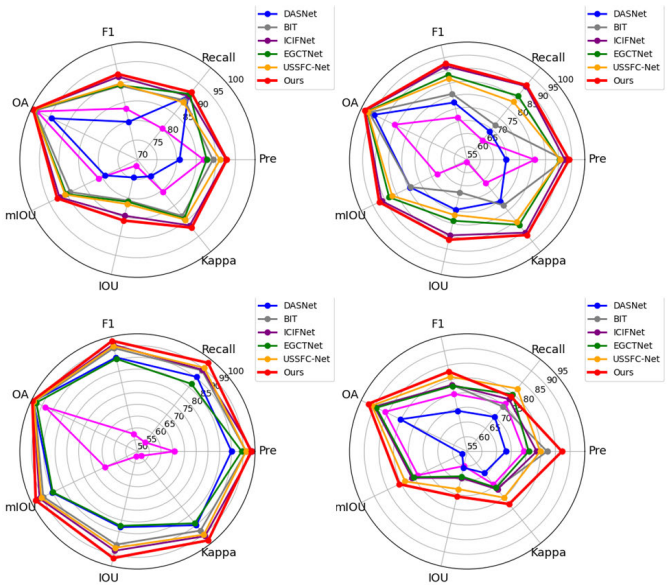


Fig. 7. Precision metric radar chart. The above figures show the accuracy metric radar charts on LEVIR-CD, Google-CD, CDD, and SYSU-CD.

To obtain a clearer view of the performance comparison between the various methods, we visualized the scores of the seven accuracy metrics using radar charts; as shown in Fig. 7, our method has several advantages in all metrics on the four datasets, and the homogeneity is good, with no poor accuracy results. Additionally, Figs. 8–11 show the visualization results of HASNet on the test sets, respectively. To clearly observe the accuracy of each area classification, we used different colors to represent TP (blue), TN (light blue), FP (red), and FN (orange).

a) Experimental results on the LEVIR-CD dataset: As shown in Fig. 8, we selected typical samples from the LEVIR-CD test set for visual comparison, such as the shadow area in the middle of the ring-shaped building in (1), the large scene changes in (2), and the strip-like small target buildings in (3). For (1), only our method shows strong robustness to shadow areas and more reasonable segmentation results for the interior and boundaries of buildings. In (2), HASNet exhibits stronger resistance to non-interesting changes and fewer misses and false detections compared to the other methods. In (3), our method not only accurately detects the complete change area of large buildings but also senses the changes in small targets at the bottom right, which are not annotated in the GT, demonstrating HASNet's strong perception of detailed information. In summary, our method achieves SOTA performance according to the qualitative results on LEVIR-CD, which is consistent with the quantitative results presented in Table II.

b) Experimental results on the Google-CD dataset: Similarly, we selected representative prediction samples from Google-CD for visual comparison. As shown in Fig. 9, large scene changes occur in (1), (2), and (3). Specifically, (1) involves interference from non-interesting changes such as roads and trees, and the challenge in (2) is how to reduce misclassifications around buildings. Despite the limited number of samples in this dataset, the DMI and our method show significant advantages in terms of detection accuracy

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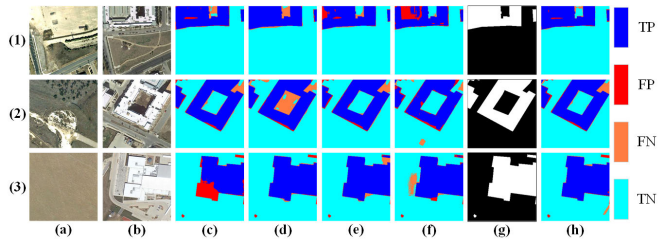


Fig. 8. Visualization of the results of several CD methods on the LEVIR-CD test set. (a) $T1$, (b) $T2$, (c) DMINet, (d) EGCTNet, (e) ICIFNet, (f) USSFC-Net, (g) GT, and (h) our method, where blue, red, orange, and light blue denote TP, FP, FN, and TN, respectively.

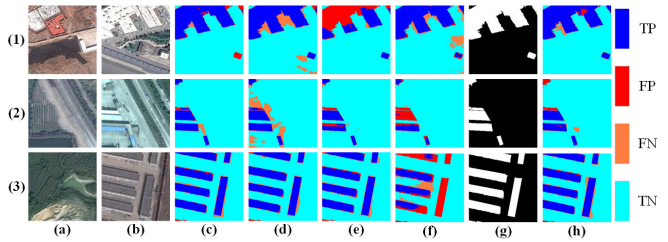


Fig. 9. Visualization of the results of several CD methods on the Google-CD test set. (a) $T1$, (b) $T2$, (c) DMINet, (d) EGCTNet, (e) ICIFNet, (f) USSFC-Net, (g) GT, and (h) our method, where blue, red, orange, and light blue denote TP, FP, FN, and TN, respectively.

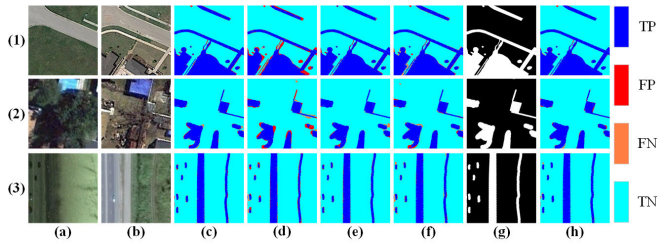


Fig. 10. Visualization of the results of several CD methods on the CDD test set. (a) $T1$, (b) $T2$, (c) DMINet, (d) EGCTNet, (e) ICIFNet, (f) USSFC-Net, (g) GT, and (h) our method, where blue, red, orange, and light blue denote TP, FP, FN, and TN, respectively.

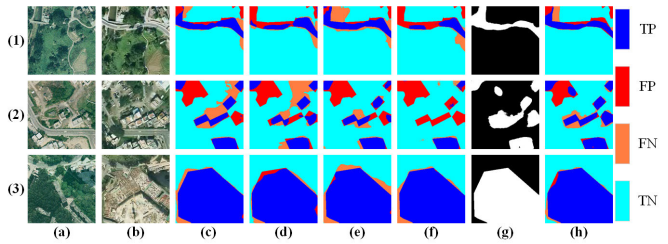


Fig. 11. Visualization of the results of several CD methods on the SYSU-CD test set. (a) $T1$, (b) $T2$, (c) DMINet, (d) EGCTNet, (e) ICIFNet, (f) USSFC-Net, (g) GT, and (h) our method, where blue, red, orange, and light blue denote TP, FP, FN, and TN, respectively.

and completeness compared to the other methods, which is consistent with the quantitative results presented in Table III. In addition, only HASNet simultaneously achieves high recognition rates and low false alarm rates for all three datasets, further demonstrating the superior performance of our method on Google-CD images.

c) Experimental results on the CDD dataset: As shown in Fig. 10, we selected typical samples from the CDD test set for visual comparison. These include the following: 1) numerous small target changes; 2) challenges due to tree occlusion affecting feature information discrimination; and 3) significant seasonal changes, which can interfere with the change detection of rural roads. As observed in Table IV, HASNet consistently achieves the highest recognition scores with the least FP and FN values, outperforming the other compared methods. This finding indicates that our method can accurately detect the changes of interest under seasonal interference, perceive small target changes, rarely miss or falsely detect changes, and is less affected by light, color, and weather.

d) Experimental results on the SYSU-CD dataset: While the first three datasets focus mainly on changes in buildings, to verify our method's detection performance for different types of target changes, we select typical samples from the SYSU-CD test set for visual comparison. As shown in Fig. 11, HASNet yields fewer misjudgments and higher recognition accuracy when detecting the changes in ambiguous areas such as roads, bare land, and residential areas; it also shows robustness and resistance to non-interesting changes such as forests and rural roads, which proves that, compared to other methods, our method shows a universal ability to discriminate the changes of interest and can more accurately detect the changes in the various categories of land objects, which is consistent with the quantitative results presented in Table V. In addition, since the SYSU-CD dataset is coarsely labeled for building areas, it does not strictly differentiate between interest samples and non-interest samples, which is detrimental to HASNet's learning of hard-case samples, resulting in a relatively low recall metric.

3) Learning Curve Comparison: To evaluate the performance of HASNet, we compared the $F1$ score variation over epochs for USSFC-Net, EGCTNet, and HASNet on four datasets. Fig. 12 shows the graphs of the evolution of the $F1$ scores for LEVIR-CD, Google-CD, CDD, and SYSU-CD from top to bottom and left to right. It is observed that our model shows higher accuracy, faster convergence, and a more stable training process compared to the other two methods, with very few large fluctuations in accuracy during training. This finding indicates that HASNet is more robust and stable than the other models. Additionally, the model converges after reaching a certain point, and training for more epochs does not significantly enhance the CD capability. Therefore, in our experiments, we set the number of epochs according to the dataset: 60 epochs for LEVIR-CD and 100 epochs for Google-CD, CDD, and SYSU-CD.

D. Ablation Study

1) Comparison of Different Loss Function Effects: To explore the contribution of EO-loss to model performance, we compared the original loss function with EO-loss. As shown in Table VI, in the control experiments using the PVT, the accuracy of the EO-loss on the three datasets is 0.18%, 1.13%, and 0.12% greater than that of the BCE and 0.11%, 0.98%, and 0.31% greater than that of the SF-loss. These results indicate that EO-loss does provide certain advantages. However, because HCSs constitute a smaller proportion of the data samples, the improvement in accuracy is not very large. This finding demonstrates the effectiveness and rationality of the method.

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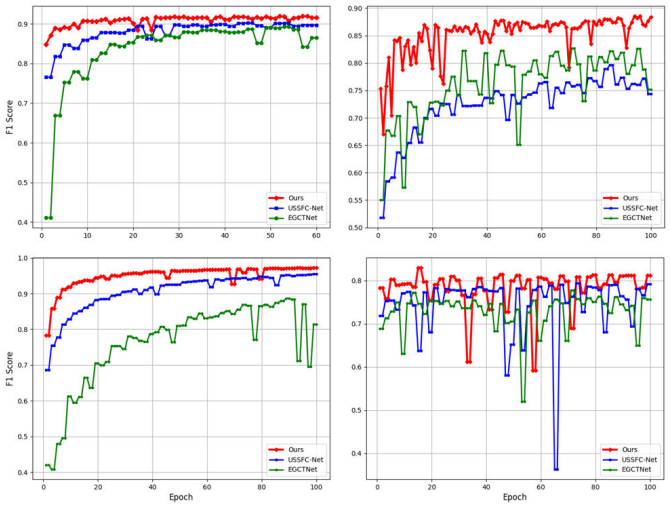


Fig. 12. $F1$ score comparison of USSFC-Net and EGCTNet on the LEVIR-CD, Google-CD, CDD, and SYSU-CD validation sets.

TABLE VI
COMPARISON OF RESULTS OBTAINED WITH DIFFERENT LOSS FUNCTIONS

Loss function	FS	Backbone		LEVIR-CD		Google-CD		SYSU-CD	
		PVT	ResNet	F1	IOU	F1	IOU	F1	IOU
BCE			✓	91.74	84.74	91.40	84.18	76.65	62.14
	✓	✓		**92.24**	**85.59**	**92.01**	**85.38**	**82.72**	**70.58**
	✓		✓	92.12	85.39	91.90	85.01	81.42	68.66
SF-loss			✓	91.77	84.80	90.87	83.27	73.39	57.96
	✓	✓		**92.31**	**85.73**	**92.16**	**85.46**	**82.53**	**70.25**
	✓		✓	91.86	84.89	91.20	83.82	82.14	69.69
EO-loss			✓	91.75	84.76	91.40	84.17	74.73	59.65
	✓	✓		**92.42**	**85.90**	**93.14**	**87.17**	**82.84**	**70.70**
	✓		✓	91.87	84.97	91.69	84.65	81.96	69.44

TABLE VII
COMPARISON OF RESULTS FOR WEIGHTING FUNCTIONS

Backbone	Norm	Linear	Exponential	Cosine	LEVIR-CD		Google-CD		CDD	
					F1	IOU	F1	IOU	F1	IOU
PVT					92.08	85.32	91.90	85.01	97.20	94.61
	✓				92.31	85.72	92.15	85.45	97.38	94.88
	✓	✓			92.23	85.59	**92.70**	**86.39**	**97.74**	**95.57**
	✓		✓		92.07	85.29	90.10	81.99	97.42	94.91
	✓			✓	**92.42**	**85.90**	92.56	86.15	97.56	95.24
ResNet					91.26	83.92	91.11	83.68	97.14	94.45
	✓				91.35	84.04	91.21	83.87	97.48	95.09
	✓	✓			91.85	84.92	91.23	83.86	**98.11**	**96.28**
	✓		✓		91.62	84.54	**92.06**	**85.28**	97.36	94.87
	✓			✓	**91.87**	**84.97**	91.69	84.65	97.85	95.78

2) Comparison of the FS-Relation Effect: To explore the contribution of the FS-relation module to HASNet performance, we conducted ablation experiments on the FS-relation module on three datasets. As shown in Table VI, when ResNet was used as the backbone network, the three sets of experiments including the FS-relation module were comprehensively superior to those without it. For example, when the loss function is BCE, the $F1$ scores of the former are 0.38%, 0.50%, and 4.77% greater than those of the latter on three datasets, and similar results are obtained for the SF-loss and EO-loss experimental sets. This finding suggests that enhancing the feature representation of hard case targets by modeling the association of foreground targets of interest with the related context in latent remote sensing spatial scene information is a feasible feature modeling strategy.

3) Comparison of Different Backbone Effects: To verify the role of the multiscale structure-designed backbone in the network, we compared the experimental results of PVT and ResNet on three datasets. As shown in Table VI, in all six sets of comparative experiments, PVT achieves more competitive accuracy results than ResNet. For example, when the loss function is EO-loss, the $F1$ scores of the PVT group are 0.55%, 1.45%, and 0.88% greater than those of the ResNet group. This finding indicates that the PVT has a stronger feature extraction capability than ResNet and is more suitable for CD tasks; this may be due to the ViT's superior global modeling capability and the multiscale information of the pyramid feature layer, which allows the FS-relation module to obtain more representative scene embedding vector SVs, thereby enhancing the feature representation of background hard case targets.

TABLE VIII
COMPARISON OF DIFFERENT FOCUSING FACTOR RESULTS

Dataset	γ	0.0	0.3	0.5	1.0	2.0	4.0	6.0
LEVIR-CD	F1	92.23	92.25	92.27	92.30	**92.42**	92.27	92.17
	IOU	85.67	85.66	85.66	85.7	**85.90**	85.65	85.49
Google-CD	F1	91.94	92.45	92.47	**93.14**	92.7	92.45	90.92
	IOU	85.09	85.97	85.99	**87.17**	86.39	85.95	83.36

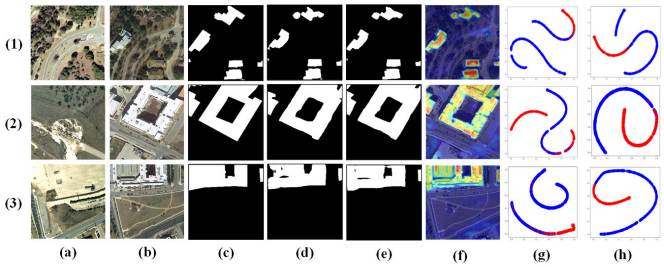


Fig. 13. LEVIR-CD. (a) $T1$. (b) $T2$. (c) GT. (d) ICIF-Net. (e) HASNet. (f) HASNet heatmap. (g) ICIF-Net t-SNE. (h) HASNet t-SNE.

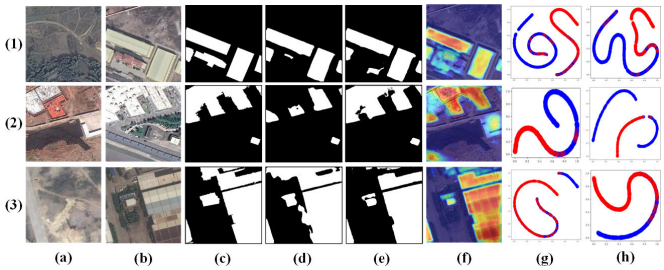


Fig. 14. t-SNE visual comparison on Google-CD. (a) $T1$. (b) $T2$. (c) GT. (d) ICIF-Net. (e) Our HASNet. (f) HASNet heatmap. (g) ICIF-Net t-SNE. (h) HASNet t-SNE.

E. Parameter Verification Experiment

There are many important parameters and functions in the proposed network, and to explore the effects of these parameters and functions on model performance, we conduct parameter validation experiments on three datasets.

1) Effect of the Dynamic Weighting Function: Optimization of HCSs in the later stages of model training is crucial for enhancing model performance. At this stage, background HCSs should become the focus of model optimization. Therefore, we introduce dynamic weight functions to shift the learning focus of the model from the foreground to the background. We designed three types of dynamic weight functions and conducted experiments on three datasets. As shown in Table VII, in the experiments using ResNet as the backbone, the maximum $F1$ score improvements achieved by using dynamic weight functions relative to the $F1$ score obtained

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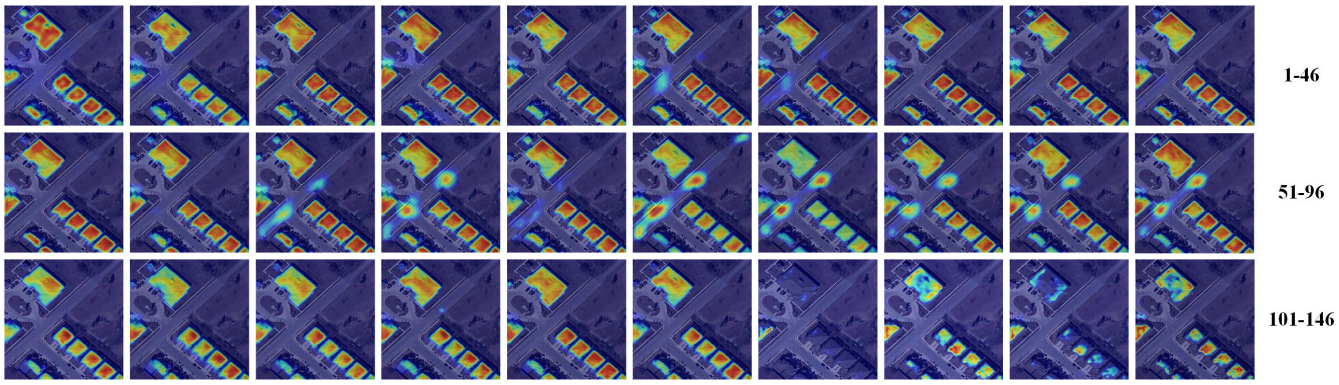


Fig. 15. Attention shift heatmap on the LEVIR-CD validation set.

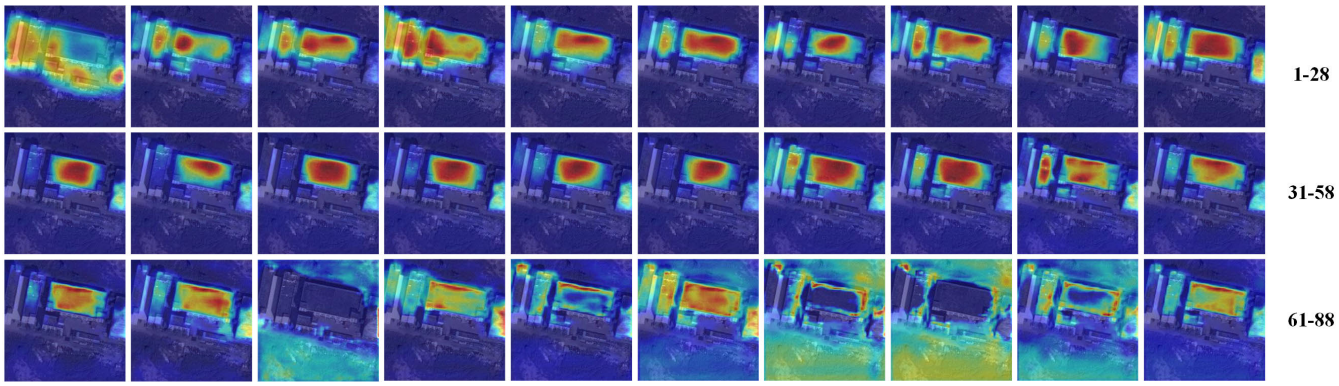


Fig. 16. Attention shift heatmap on the Google-CD validation set.

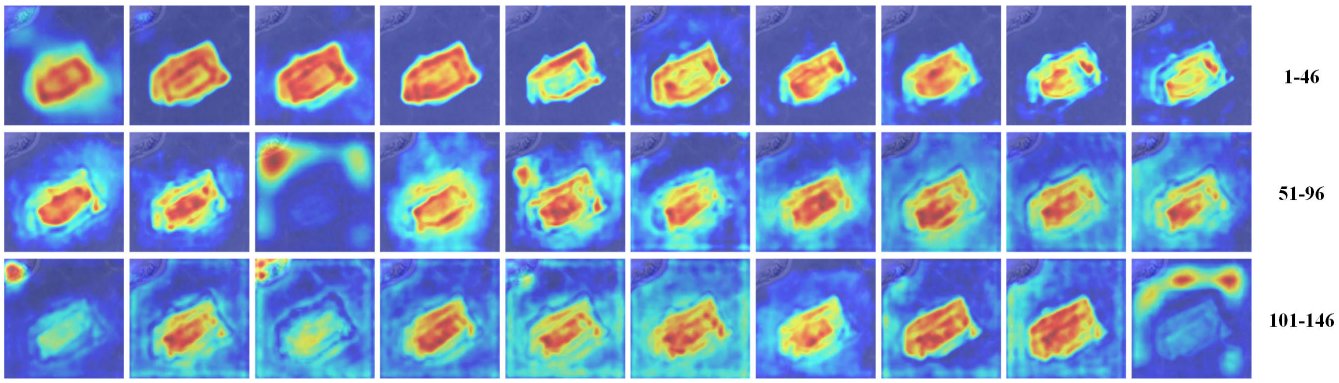


Fig. 17. Attention shift heatmap on the CDD validation set.

without the use of these functions were 0.52%, 0.85%, and 0.63%, respectively; similar results were obtained in the experiments using PVT as the backbone, with increases of 0.34%, 0.80%, and 0.54%, respectively. This finding indicates that the method based on dynamic weighting reduces early training errors in predicting HCSs, thereby enhancing the CD performance of the model; moreover, focusing on the optimization of background HCSs in the later stages of training is both reasonable and effective. According to both sets of experiments, although all three dynamic weight functions achieved some accuracy improvements, they had no absolute advantage; this may be due to the varying numbers and distributions of HCSs in different datasets; hence, different dynamic weight functions are suitable for different datasets. We can choose the appropriate dynamic weight function on the basis of data characteristics. For example, cosine dynamic weighting is more suitable for datasets with a clear gap between simple and HCSs, allowing for stable adjustment of the loss distribution to achieve robust convergence; exponential dynamic weighting is more suitable for datasets with fewer foreground samples and more background samples.

2) Effect of the Focusing Factor: The hard case awareness term $(1-p)^{\gamma}$ leverages the distribution of loss values to approximate the regions of HCSs and then adjusts the attention weight distribution between HCSs and foreground samples using the focus factor γ (shown as Table VIII). Generally, a larger γ corresponds to a greater weight and a higher degree of attention. We conducted comparative experiments with progressively increasing γ on two datasets and observed similar experimental effects. As shown in Table VIII, starting from $\gamma = 0.0$, the CD accuracy continuously improved with increasing γ. For the LEVIR-CD and Google-CD datasets, when γ was set to 2.0 and 1.0, respectively, the $F1$ scores were 0.19% and 1.20% higher than when γ was 0.0, reach-

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ing absolute accuracies of 92.42% and 93.14%, respectively. However, as γ continues to increase, the performance of the model began to decline; this may be due to two reasons: first, image noise was also given a high focus factor γ, leading to its misclassification as HCSs; second, an excessively large γ caused the model to excessively neglect foreground targets, thus reducing its discriminative ability for foreground interest targets. Based on these results, we can conclude that an appropriate focus factor γ can enhance the model's CD performance, but different data characteristics often require different optimal focus factors.

F. Analysis of Effectiveness

1) t-SNE and Attention Heatmap Visualization Effect: To demonstrate how our method focuses on various remote sensing targets, we visualized the attention heatmap of the last layer of HASNet and overlaid it on the $T2$ image to obtain the image shown in (f). In (f), the attention distribution of the changing targets can be clearly observed, where red denotes higher attention values and blue denotes lower values. As shown in Figs. 13(f) and 14(f), areas with high heating values completely cover the areas of interest changes. The house change areas in LEVIR-CD and Google-CD both have high heating values, indicating that our method learns the feature representation of interest targets and is not sensitive to non-interesting changes. Notably, HASNet demonstrates strong robustness to building shadows, as shown in (2) and (3) of Fig. 13. The shadows around buildings do not affect our method's ability to accurately detect real change areas. Moreover, shadows around buildings may be a key factor in change recognition because our model can implicitly learn certain additional feature concepts to facilitate the identification of change areas.

In addition, to more intuitively reflect the degree of separation between the changed and unchanged samples, we used t-distributed stochastic neighbor embedding (t-SNE) [58] to downscale the distribution of the visualized samples and compare and analyze our method with the distribution of the ICIF-Net features. As shown in (g) and (h) of Figs. 14 and 15, red represents changed samples, and blue represents unchanged samples. Notably, in the dimensionality reduction results of HASNet, the separation boundary between the red and blue samples is clearer than the boundaries between the other samples, with both sample categories being more distinctly clustered and with few instances of red and blue intermingling. This finding indicates that our method has more accurate change perception and anti-interference capabilities, leading to the greater change confidence and clear boundaries of HASNet. By contrast, in the ICIF-Net dimensionality reduction results, a certain number of red samples appear in the blue samples, indicating that some of the changed samples and the unchanged samples are indistinguishable in the feature space; this leads to some samples at the edge of the changed area with low confidence being misclassified by the network, resulting in "blurred boundaries of the changed area."

2) Attention Shift Heatmap Visualization: To validate the attention shift pattern of the model during the early and late training stages, we visualized the attention focus distribution on sample data at different epochs. Each epoch is equidistant, allowing the observation of the pattern of shifting optimization focus. Figs. 15–17 show the heatmaps of a particular data sample in LEVIR-CD, Google-CD, and CDD, respectively, visualized using Grad-CAM [59]. All three images exhibit the same pattern: in the early stages of training; the optimization focus is concentrated on the interest targets in the foreground, which in this case are "buildings." By the middle stages of training, the model gradually shifts toward learning about background information, such as the areas around roads or less obvious buildings. In the later stages of training, the model's attention increasingly turns to learning about background features, inducing the neglect of foreground targets to various degrees.

V. Conclusion

To address the challenge of obtaining HCSs from remote sensing imagery, this article presents a Siamese foreground association-driven HCS optimization network, named HASNet that includes a foreground-scene association module and an EO-loss. In the EO-loss function, the distribution of HCSs is estimated through the hard case awareness term, and the optimization focus of the network is shifted dynamically to balance the learning process for both foreground targets and background HCSs. In the foreground-scene association module, the associations between interest targets in the foreground and related context in latent remote sensing spatial scene information are modeled, and background HCSs are actively output to ensure the successful extraction of interest changes in both foreground and background HCSs. This strategy effectively captures and perceives changing features, showing advantages in detecting the changes in HCSs. The experimental results on four change detection datasets show that, compared to the other methods, our method obtains more accurate CD results, is less prone to problems such as omissions and misdetections, and is more robust against HCSs.

Although it is applied only to CD tasks here, the proposed strategy can be readily transferred to other application scenarios. Of course, supervised learning algorithms are limited by data availability. Therefore, in the future, we plan to further explore optimization methods for HCSs in semi-supervised or self-supervised learning [60], [61], [62] contexts and with causality considered [63], [64].

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